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Superseding
MIL-M-81850(AS)
14 June 1972

MILITARY SPECIFICATION

MAGNETIC TAPE TRANSPORT RD-319A/AYA-8

This Specification has been approved by the
Naval Air Systems Command, Department of the Navy

1. SCOPE

1.1 Scope - The equipment covered by this specification shall record digital data on and read digital data from magnetic tapes under control of the data analysis logic unit MX-8024/AYA-8 (Logic Unit 2).

1.2 Classification - The equipment covered by this specification shall consist of a Magnetic Tape Transport (MTT) with the following subsystems:

1.2.1	Power System
1.2.2	Interface System
1.2.3	Control System
1.2.4	Capstan Drive System
1.2.5	Reel Drive System
1.2.6	Data System
1.2.7	Pneumatic System
1.2.8	Diagnostic System

1.3 Associated Equipment - The equipment shall operate in conjunction with the associated equipment listed in 6.7.

2. APPLICABLE DOCUMENTS

*2.1 General - The following documents form a part of this specification to the extent specified herein. Listed are the issues of the documents in effect for the previous procurement. However, in lieu of the issue listed, the contractor shall use the latest issue in effect of these documents where feasible. If the use of the latest issue will affect design performance, or interchangeability of any replaceable part, then the issue of the document listed below shall be used.

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SPECIFICATIONS

Federal

W-T-0051B	Tape, Electronic data processing, (Magnetic oxide-coated, ½ inch)

Military

MIL-E-5400H 1 June 1965	Electronic Equipment, aircraft, General Specification for
MIL-T-5422E 15 November 1961	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181D 26 November 1959	Interference Control Requirements, Aircraft Equipment
MIL-M-7793C (Amendment #1) 22 May 1962	Meter, Time Totalizing
MIL-N-18307C 1 October 1958	Nomenclature and Nameplates for Aeronautical Electronics and Associated Equipment
MIL-E-17555F 5 March 1965	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-S-19500E 1 September 1966	Semi-Conductor Devices, General Specification
MIL-T-18303B 1 April 1968	Test Procedures, Preproduction and Inspection, for Aircraft Electronic Equipment
MIL-C-26482D 10 May 1966	Connectors, Electric, Circular, Miniature Quick Disconnect

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MIL-C-81511	Connectors, Electrical, Circular, High Density, Quick Disconnect, Environment Resisting, General Specifications for

STANDARDS

Military

MIL-STD-701G 9 October 1967	Preferred and Guidance List of Semiconductor Devices
MIL-STD-704A 15 May 1963	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781A 15 November 1967	Reliability Tests, Exponential Distribution
MIL-STD-785 30 June 1965	Requirements for Reliability Program (for System and Equipment)
MS-17322	Meter, Time Totalizing, 115 Volt - 400 cycle

Naval Air Systems Command

EI-515	Avionics Installation Instructions for Data Analysis Programming Group AN/AYA-8
AR-34 26 March 1969	Failure Classification for Reliability Testing, General Requirement for

OTHER PUBLICATIONS

WR-101 Part I 28 February 1966 (Amendment 2)	Electromagnetic Control Requirements for Advanced ASW Avionics Systems; A-NEW
WR-101 Part II 3 March 1966 (Amendment 2)	Selection and Testing of Microelectronic Circuitry for Advanced ASW Avionics Systems; A-NEW

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2.1.1 Availability of Documents

(1) When requesting specifications, standards, drawings and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Publications and Forms Center, Code 105 Tabor Avenue, Philadelphia, Pennsylvania 19120.

3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

3.2 Parts and Materials - In the selection of parts and materials fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern.

(1) Microelectronic Technology shall be considered and microelectronic items shall conform to requirements specified in WR-101, Part II.

(2) Other parts and materials requirements shall conform to Specification MIL-E-5400.

(3) Nonrepairable subassemblies, as outlined in Specification MIL-E-5400, shall be used when practicable. Nonrepairable subassemblies must be reliable.

(4) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

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3.2.1 Nonstandard Parts and Materials Approval- Approval for the use of nonstandard parts and materials (including electron tubes, transistors and diodes) other than microelectronic devices shall be obtained as outlined in Specification MIL-E-5400. Microelectronic devices shall be approved as outlined in WR-101, Part II, Amendment 1.

3.2.2 Modules - The electronic portions of the equipment shall be functionally modularized in accordance with paragraph 3.3.7.2 of this specification.

3.3 Design and Construction - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

*3.3.1 Total Weight - The total weight of the equipment shall be a minimum consistent with good design and shall not exceed 240 pounds, exclusive of magnetic tape weight.

3.3.2 Reliability - The contractor shall conduct a reliability program in accordance with MIL-STD-785. On a re-order from a supplier who has previously produced the equipment, the program previously used may be continued unless otherwise indicated in the contract or order.

3.3.2.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently for a period of at least 400 hours without the necessity for readjustment of any controls or mechanical adjustments which are inaccessible to the operator during normal use.

3.3.2.2 Operating Life - The equipment shall have a total operating life of 50,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

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3.3.2.3 Reliability in Mean Time Between Failure (MTBF) - The equipment, including any built-in test provisions, shall have 650 hours of mean (operating) time between failures when tested in accordance with 4.4.2 of this specification.

3.3.3 Cabling and Connectors

3.3.3.1 Cables - The equipment shall provide for the use of cables in accordance with Specification WR-101 Part I.

3.3.3.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification WR-101 Part I, except as modified herein, and the detailed requirements of EI-515. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

3.3.4 Control Panel - The control panel shall be attached to the equipment with all controls and indicators required for normal operation available to an operator.

3.3.5 Interchangeability - The equipment shall meet the interchangeability requirements of Specification MIL-E-5400.

3.3.6 Interference Control - The equipment shall conform with the requirements of WR-101, Part I for design, construction and workmanship, except as specified in the following limits:

1. Conducted Interference on Power Lines
   Broadband Interference
   Allowable Limit: Reference Figure 3, Appendix D of WR-101 Part I
   (1) The limit from 30 Hz to 15 KHz is increased from 100 dbua/ 20 KHz to 140 dbua/ 20 KHz.

   (2) A straight line from 15 KHz and 176db greater than one microamp/MHz to 30 KHz and 170 db greater than one microamp/MHz.

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(3) A straight line from 30 KHz and 170 db greater than one microamp/MHz to 2 MHz and 50 db greater than one microamp/MHz.

(4) The limit from 2 MHz to 25 MHz remains the same as shown in figure 3.

II. Radiated Interference

Broadband Interference

Allowable Limit: Reference Figure 5, Appendix D of WR-101, Part I.

(1) A straight line from 15 KHz and 125 db greater than one microvolt/MHz to 40 KHz and 114 db greater than one microvolt/MHz.

(2) A straight line from 40 KHz and 114 db greater than one microvolt/MHz to 150 KHz and 77 db greater than one microvolt/MHz.

Broadband Interference: Reference Figure 16 of MIL-I-6181D.

(1) The limit from 150 KHz to 25 MHz remains as shown in Figure 16.

(2) A straight line from 25 MHz and 54 db greater than one microvolt/MHz to 300 MHz and 72 db greater than one microbolt/MHz.

(3) A straight line from 300 MHz and 72 db greater than one microvolt/MHz to 1000 MHz and 65 db greater than one microvolt/MHz.

Provisions for Maintainability

3.3.7.1 Maintainability, General - The equipment shall be designed in order to facilitate fault detection, isolation and repair

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action withoug removing the equipment from its normal operational environment in the aircraft or removing connectors. The use of special purpose test equipment shall not be permitted. Any special test requirements shall be included in the design of the equipment.

3.3.7.2 Maintainability Objectives - Fault isolation shall be automatic using a computer initiated diagnostic program. Where computer programming is not feasible, the equipment shall provide readily accessible test points to accommodate standard test equipment. Fault isolation shall be to the least complex segment of the equipment which is capable of being removed and reinserted within the equipment through the use of a connector rather than by soldering (brazing) or welding defined here as a maintenance module. Isolation to and replacement of failed modules shall be accomplished in 15 minutes or less for 50% of such failures and in a mean time of less than 30 minutes for 95% of all failures.

3.3.7.3.1 Diagnostic Tests - The Diagnostic Program shall utilize the computer to diagnose MTT equipment malfunctions (excluding power supply). The Diagnostic Program shall be capable of delineating an MTT failure to the failed module to the extent of the interface provided to the diagnostic program. To the extent permitted by the characteristics of the MTT, the Diagnostic Program shall satisfy the following requirements:

(1) Comprehensive - Every functional element shall be tested in all of its functions within the limits of the Diagnostic Hardware constraint.

(2) Automatic - Operator intervention shall be held to a minimum. To satisfy this requirement, test loops should be included in the design of the MTT. The test loops shall permit the computer to selectively monitor the performance of the functional elements in the MTT for purpose of fault detection and isolation. The criterion to be used in the design of the test loops is as follows. The amount of hardware for the test loops shall not exceed 15% of the total hardware for the unit.

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3.3.7.4 Maintainability- The equipment shall consist of readily removable and replaceable subassemblies to insure that the diagnostic, maintenance and repair provisions of this specification will be met.

Captive Hardware - Captive hardware shall be used in securing access panels and any removable printed wiring board assemblies.

3.3.7.6 Test Points - Electrical test points designed to accept a .080" diameter probe shall be provided on printed wiring boards to assist in fault isolation without requiring removal of the board from the equipment.

3.3.7.7 MTT Duty Cycle - A minumum duty cycle of 0.50 shall be used for tests of extended duration. Duty cycle is defined as the ratio of tape motion time to power on time, with at least 1000 start/stop transitions per hour of tape motion. A duty cycle of 0.20 shall be used for maintenance purposes.

3.3.7.8 Special Tools - The number of special tools required for the MTT shall be kept to a minimum. For tools that are required, two sets shall be provided with each MTT.

3.3.8 Nomenclature, Nameplates and Identification Marking - Nomenclature and serial number assignment, nameplate approval and identification marking shall be in accordance with Specification MIL-N-18307.

3.3.9 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room Ambient (25°C±5°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input Power	115±1.0 VAC, 3 phase 400 Hz
Voltage	±5%
Frequency	±20 Hz

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3.3.10 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class 1A equipment, except as modified herein.

3.3.10.1 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of Curves I and II of Specification MIL-E-5400 or ±2 g whichever is less.

3.3.10.2 Temperature - The equipment shall operate satisfactorily over the temperature range specified in MIL-E-5400 except that from -54°C to -25°C it is only necessary that the equipment shall not be damaged by application or removal of input power and except frost test shall not be performed.

3.3.10.3 Altitude - The equipment shall operate satisfactorily over an altitude range of 0 to 15,000 ft. Between 15,000 and 20,000 ft. no damage shall occur to the equipment and/or magnetic tape contained on the transport; and information contained on the magnetic tape shall not be modified or destroyed.

3.3.10.4 Attitude - The equipment shall operate satisfactorily in all attitudes and orientations within ±90° of its normal mounting orientations. The normal mounting orientation shall be with the tape reels in the vertical plane.

3.3.10.5 Explosion - Explosion proofing is not required.

3.3.10.6 Sand And Dust Proofing - Not required.

3.3.11 Warm-Up-Time - The time required for the equipment to warm up prior to operation shall be kept to a minimum and shall not exceed the following limits:

Ambient Temperature	Time
-25°C to 0°C	5 minutes maximum
0°C to +55°C	30 seconds maximum

3.3.12 Input Electrical Power

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3.3.12.1 Operating Power - The equipment shall meet all applicable requirements of MIL-STD-704A and shall give specified performance when energized from power sources having characteristics as defined in MIL-STD-704A and limits as modified herein. The power required shall not exceed the following amounts:

(1) AC Power (Three Phase), 115/200V, Category B, (1400 watts) 1900 VA.

3.3.12.1.1 Normal Operation - The MTT shall perform all functions programmed by the Magnetic Tape Control (MTC) of Logic Unit 2 within the specified limits of this document under the following conditions.

3.3.12.1.1.1 Reduced Performance - Input voltages between 102 to 108 and 118 to 124 VAC shall constitute a reduced performance environment. Reference paragraph 3.5.1.3.6.2.4.3.

3.3.12.1.2 AC Voltage Transients - The input voltage transient limits for normal operation are shown in a graph in Figure 3 (attached) and shall be as follows:

I. The high limit for specified normal operation shall be as follows with reference to Figure 3 of MIL-STD-704A.

(1) A straight line from 500 microseconds and 160 volts (coinciding with curve 2) to 28 milliseconds and 160 volts.

(2) A straight line from 28 milliseconds and 160 volts to 0.26 second and 124 volts.

(3) A straight line from 0.26 seconds and 124 volts to 1.0 second and 118 volts at which point the limit 2 curve is joined for the remainder of the graph.

II. The low limit for specified normal operation shall be as follows with reference to Figure 3 of MIL-STD-704A.

(1) [illegible] from 500 microseconds and 60 volts to 5.8 millisec[illegible] and 60 [illegible]

(2) A straight line from 5.8 milliseconds and 60 volts to 70 milliseconds and 108 volts.

(3) A straight line from 70 milliseconds and 108 volts to 1 second and 108 volts at which point curve 3 is joined for the remainder of the graph.

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3.3.12.1.3 AC Overvoltage Protection - The equipment shall not be damaged by voltage above Limit 2 but less than Limit 1 of Figure 3 of MIL-STD-704 and shall automatically return to the ready condition when the voltage returns within normal operating limits.

3.3.12.1.4 AC Undervoltage Protection - The equipment shall not be damaged by voltages below 80 volts for any duration and shall automatically return to the ready condition when the voltage returns within normal operating limits.

3.3.12.1.5 Overload Protection - Overload protection for the equipment shall be provided internally. All parts and circuits which are likely to carry an overload due to any failures shall be proportioned to withstand such overload without permanent damage to the equipment or shall have suitable protective devices. Circuit breakers shall be used when possible.

3.3.12.1.6 Loss of Power - No damage shall result from accidental or deliberate stoppage of electrical power to the equipment regardless of the time in its operating cycle that such a stoppage shall occur, and regardless of the time interval that elapses before power is restored. The equipment shall enter the "Ready" condition (i.e., operable but not performing a specified operation) and provide a Status Error Indication to the MTC whenever power is restored. It shall be the function of the Computer Program or the manual operator to insure the proper sequence of starting instructions to continue normal operation. The MTT shall not write data nor alter existing data as a result of MTT AC Power removal.

3.3.12.1.7 MTT Initiation - The MTT shall be self-initiating, when power is applied and the MTT power switch is in the ON position, no spurious data shall be transmitted to the MTC or written on the tape and the MTT shall be ready for normal operation.

3.3.12.2 Power System - The equipment shall contain its own power system which meets the requirements of MIL-STD-704 as modified herein.

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3.3.12.2.1 Power Control Switch - A power control switch that controls the electric power to the equipment shall be mounted on the control panel of the equipment.

3.3.13 Cooling Requirements - The equipment shall be designed to insure adequate cooling of all components and assemblies over the ambient temperature range dictated by MIL-E-5400 for Class 1A equipment as modified by para 3.3.10.2.

3.3.14 Blowers - The use of blowers is permitted to meet the cooling requirements of paragraph 3.3.13.

3.4 Performance - Unless otherwise specified, values set forth to establish the requirements of satisfactory performance apply to performance under both standard and extreme service and input power conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.4.1 Operation - The equipment shall be designed to transport a magnetic tape past a seven channel read/write erase head under control of the MTC and perform the following functions:

(1) Receive information from the MTC and record it on magnetic tape.

(2) Read data from the magnetic tape and transmit it to the MTC.

(3) Receive and store instruction signals from the MTC and modify its own operation or change its condition in accordance with the signals received.

3.5 Detail Requirements

3.5.1 Magnetic Tape Transport (MTT) RD-319A/AYA-8

3.5.1.1 Function - The equipment is a fixed-speed airborne tape recorder/reproducer that performs fast start/stop, digital read

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and write operations. Its function is to record or reproduce (write/read) seven channels of digital information. The read function is performed while moving tape in either direction. The write function is performed only while moving tape in the forward direction. A bi-directional tape speed of 75 inches per second (ips) and a fast rewind speed is provided. During normal operation the MTT is controlled by logic commands from Logic Unit 2 of the data processing system (DPS). When not under remote control, the MTT logic control panel pushbuttons may be used to control tape movement. The write/read function is disabled under local control. The MTT local control panel indicators function during local and remote operation. The erase function may be manually generated during local control.

3.5.1.2 Form Factor - The total volume shall be no greater than 4.0 cubic feet, not including the mounting flanges and the handles mounted thereto. The overall dimensions shall not exceed 26 inches in height, 30 inches in width and 10.8 inches in depth.

3.5.1.3 Contents - The Magnetic Tape Transport shall contain the following functional systems:

3.5.1.3.1 Power System

3.5.1.3.1.1 Functional Description - The power system consists of the power supply subsystem and the power control unit. The power supply subsystem uses AC input power to produce all required DC voltages for use in the various electrical systems within the equipment. The power control unit provides AC input power control, distribution and protection. It monitors the AC input line voltage and provides automatic equipment shutdown when an overvoltage or undervoltage is sensed. Additionally, the power control unit monitors each DC supply voltage and provides equipment shutdown for over-voltage or undervoltage conditions and for short circuits on any supply output.

3.5.1.3.2 Interface System

3.5.1.3.2.1 Functional Description - The Interface System receives all input commands, digital data to be recorded on magnetic tape, and diagnostic signal inputs from the Magnetic Tape Control (MTC). All outputs from the MTT are passed through the Interface System and transmitted to the MTC.

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3.5.1.3.2.2 Interface Requirements

3.5.1.3.2.2.1 Magnetic Tape Control to Magnetic Tape Transport

3.5.1.3.2.2.1.1 General

(1) All communications between the MTC and the MTT shall be via twisted pairs and data line driver/input amplifier combination, specified in Appendix I unless otherwise noted.

(2) A logical "1" shall be 0+0.5-0.0 volts and a logical "0" shall be 4±1 volts, unless otherwise noted.

(3) The MTT shall gate all signals with the select line, unless otherwise noted.

3.5.1.3.2.2.1.1.2 Command Interval - The time from the termination of one command to the initiation of a new command shall be 6 milliseconds minimum.

3.5.1.3.2.2.1.1.3 Seven Data Lines - Seven Data Lines, indicating the information to be written on the tape, shall be transmitted to the MTT.

3.5.1.3.2.2.1.1.4 Data Strobe - A Data Strobe line, indicating valid data to the MTT shall be transmitted. The Data Strobe shall be a $2.0^{+1.0}_{-0}$ microsecond pulse: The data lines shall be stable at the initiation of and shall remain stable for the duration of the pulse and for 2 microseconds (min) after the Data Strobe pulse.

3.5.3.2.2.1.1.5 Select Line - One select line shall be transmitted to the MTT indicating the computer selection. The Select line shall be a logical "1" as long as the MTT is selected.

3.5.1.3.2.2.1.1.6 Forward - A forward line, indicating to the MTT that it shall move the tape in the forward direction, shall be transmitted. The Forward line shall be logical "1" as long as the tape is to travel in the forward direction.

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3.5.1.3.2.2.1.1.7 Reverse - A reverse line, indicating to the MTT that it shall move the tape in the reverse direction, shall be transmitted. The Reverse line shall be a logical "1" as long as the tape is to travel in the reverse direction.

3.5.1.3.2.2.1.1.8 Rewind - A rewind signal, indicating to the MTT that it shall rewind the tape at 150 inches per second to the load point, shall be transmitted. The Rewind signal shall be a logical "1" for 2.0+2.0-0 microseconds.

3.5.1.3.2.2.1.1.9 Read - A Read line, indicating to the MTT that it shall read data from the tape, shall be transmitted. The Read line shall be a logical "1" as long as the MTT is to read data.

3.5.1.3.2.2.1.1.10 Write - A write line, indicating to the MTT that it shall write data onto the tape, shall be transmitted. The Write line shall be a logical "1" as long as the MTT is to write data and shall occur only when the MTT is to move tape in a forward direction.

3.5.1.3.2.2.1.1.11 Write Reset - A write reset signal, indicating to the MTT that it shall reset its write register, shall be transmitted. The write reset signal shall be a logical "1" for a 2.0 microseconds (min.).

3.5.1.3.2.2.1.1.12 Three Packing Density - Three Packing Density lines, indicating to the MTT a packing density of 800,555.5 or 200 bits per inch, shall be transmitted. Only one packing density line shall be a logical "1" at any one time: However, the chosen Packing Density line shall remain a logical "1" for as long as the function being performed is active.

3.5.1.3.2.2.1.1.13 Two Address Lines - Two Address Lines, indicating that the address of a given MTT is to be as manually selected from Logic Unit 2 shall be transmitted to each MTT. Only one Address line shall be a logical "1" at any one time and shall remain active until some manual intervention occurs.

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3.5.1.3.2.2.1.1.14 Interlock - Two interlock lines shall be provided to the MTT. Continuity shall exist between the two lines only while MTC power is turned on and the MTC subassembly which provides motion commands to the MTT is in its normal operating position.

3.5.1.3.2.2.1.1.15 MTT Test Enables (Five Lines) - When an MTT Test Enable Line is a logical "1" that test mode shall be enabled.

3.5.1.3.2.2.1.1.16 Master Clear - A Master Clear Line shall be transmitted to the MTT indicating that all motion commands should be terminated. The Master Clear signal shall be a logical "1" for 2.0 microseconds (min.).

3.5.1.3.2.2.2 Magnetic Tape Transport to Magnetic Tape Control

3.5.1.3.2.2.2.1 General

(1) All communications between the MTT and the MTC shall be via twisted pairs and data line driver/input amplifier combinations unless otherwise noted. See Appendix I for Input Amplifier Specifications.

(2) A logical "1" shall be 0+0.5-0.0 volts and a logical "0" shall be plus 4±1 volts, unless otherwise noted.

3.5.1.3.2.2.2.2 Seven Data Lines - Seven Data Lines, indicating the information read from the tape, shall be transmitted from each MTT to the MTC.

3.5.1.3.2.2.2.3 Data Strobe - A Data Strobe line, indicating valid data to the MTC shall be transmitted from the MTT to the MTC. The Data Strobe shall be a 2.0 microsecond (min.) pulse: the data lines shall remain stable for the duration of pulse and for 2 microseconds after the Data Strobe pulse.

3.5.1.3.2.2.2.4 Select Acknowledge - One select acknowledge line, indicating that the MTT has recognized the Select line, shall be transmitted from MTT to MTC. The Select Acknowledge line shall be a logical

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"1" as long as the MTT is selected. If two cables are used for the MTC/MTT interface, the Select signal and the Select Acknowledge Signal shall not be in the same cable.

3.5.1.3.2.2.2.5 Load Point - One Load Point line, indicating that the tape is at the load point, shall be transmitted from the MTT to the MTC. The Load Point line shall be a logical "1" as long as the tape is at load point.

3.5.1.3.2.2.2.6 End of Tape Warning - A continuous End of Tape Warning indicating that the tape marker has passed the end of tape sensor, shall be transmitted from the MTT to the MTC.

3.5.1.3.2.2.2.7 Ready - One Ready line, indicating that the MTT is not in a fault condition, shall be transmitted from the MTT to the MTC. The ready line shall be a logical "1" as long as the MTT is in the Ready condition.

3.5.1.3.2.2.2.8 Rewinding - One rewinding line, indicating that the tape is rewinding shall be transmitted from the MTT to the MTC. The Rewinding line shall be a logical "1" as long as the MTT is rewinding tape and until tape is positioned at load point or a master clear from the MTC is received.

3.5.1.3.2.2.2.9 Write Lockout - One Write Lockout line, indicating that the tape supply reel does not contain a write permit ring, shall be transmitted from the MTT to the MTC. The Write Lockout line shall be a logical "1" as long as the supply reel does not contain a write permit ring.

3.5.1.3.2.2.2.10 Low Tape - One Low Tape line indicating that the tape is at 75 ± 25 feet remaining shall be transmitted from the MTT to the MTC. The Low Tape line shall be a logical "1" as long as the tape remaining is less than preset amount.

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3.5.1.3.3 Control System

3.5.1.3.3.1 Functional Description - The Control System consists of the Control Panel, the Control Logic Unit, and the Low Tape Control. The Control Logic monitors the status of the MTT and generates signals to enable and control all modes of tape motion. The MTT is normally controlled remotely from the MTC through the interface system described in paragraph 3.5.1.3.2. When the MTT is not selected for remote operation, control is automatically transferred to the control panel.

3.5.1.3.3.2 Operating Requirements

3.5.1.3.3.2.1 The MTT shall contain circuits to receive and store instruction signals from the MTC and circuits to modify its own operation or change its condition by returning status signals identified in paragraph 3.5.1.3.2.2.2 to the MTC.

3.5.1.3.3.3 Control Panel

3.5.1.3.3.3.1 General - The Control Panel shall be attached to the MTT with controls, indicators or adjustments available to an operator. Refer to EI-515, Avionics Installation Instruction for Data Analysis Programming Group AN/AYA-8 for Control Panel Layout details.

3.5.1.3.3.3.2 Indicator and Control Switches - The following indicators and controls shall be mounted on the front of each MTT:

(1) MTT Address Number - Shall display the assignment number assigned to the MTT, Address number shall be either #1 or #2.

(2) Write - Shall indicate when a write operation is initiated.

(3) Read - Shall indicate when a read operation is initiated.

(4) Write Lockout - Shall indicate the Supply Reel does not contain a write ring.

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(5) Forward - Shall indicate that forward tape motion was conditioned. Pressing the switch shall condition the MTT for forward operation.

(6) Reverse - Shall indicate that reverse tape motion was conditioned. Pressing the switch shall condition the MTT for reverse operation.

(7) Rewind - Shall indicate that the MTT is rewinding tape. Pressing the switch shall initiate tape rewind only if the reverse condition was previously set by the reverse switch.

(8) Tape Motion Switch - Shall start tape motion in the direction conditioned previously by the forward or reverse switch.

(9) End of Tape - Shall indicate either the "End of Tape" or a tape break.

(10) On/Off - Shall turn the MTT unit on if it was off, and off if it was on.

(11) Low Tape Warning - Shall display to the operator whenever the tape remaining on the supply reel is less than 75 ± 25 feet remaining.

3.5.1.3.4 Capstan Drive System

3.5.1.3.4.1 Functional Description - The Capstan Drive System initiates and regulates the movement of the magnetic tape across the magnetic head under servo control on the mode commanded by the Control System.

3.5.1.3.4.2 Operational Requirements

3.5.1.3.4.2.1 Speed - The tape speed shall be 75 inches per second ± 3% for all operating conditions except rewind.

3.5.1.3.4.2.2 Direction - The MTT shall be capable of bi-directional operation.

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3.5.1.3.4.2.3 High Speed Rewind - The tape speed in rewind shall be 150 inches per second minimum.

3.5.1.3.4.2.4 Start-Stop Characteristic Requirements - The combined start time, start distance, stop time and stop distance characteristics of the MTT shall conform to the requirements shown in this specification.

3.5.1.3.4.2.5 Start-Stop Characteristic Definitions -

3.5.1.3.4.2.5.1 Start Time - Start time is defined as the time measured from the initiation of a motion command for the tape under the READ/WRITE head, to achieve a speed of 72.75 IPS.

3.5.1.3.4.2.5.2 Start Distance - Start distance is defined as the tape motion which takes place during start time and shall not exceed .350 inches.

3.5.1.3.4.2.5.3 Stop Time

3.5.1.3.4.2.5.3.1 Stop Time to 9.375 IPS - is defined as the time from termination of a motion command for the speed of the capstan to be less than 1/8 normal.

3.5.1.3.4.2.5.3.2 Full Stop Time - Stop time is defined as the time, measured from the termination of a motion command, for the tape under the READ/WRITE head, to cease all motion.

3.5.1.3.4.2.5.4 Stop Distance - Stop distance is defined as the tape motion which takes place during stop time and shall not exceed .350 inches.

3.5.1.3.4.2.6 Maximum Start Time - The start time shall not exceed 5.87 milliseconds.

3.5.1.3.4.2.7 Maximum Stop Time

3.5.1.3.4.2.7.1 Time to 9.375 IPS - shall not exceed 4.5 milliseconds.

3.5.1.3.4.2.7.2 Full Stop Time - The stop time shall not exceed 25.0 milliseconds.

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3.5.1.3.4.2.8 Tape Creep - That portion of the tape in contact with the read/write head shall not move after a stop function has been completed unless a new command is given. In addition, no capstan creep shall occur when primary power is applied to or removed from the MTT with the MTT in a stopped condition.

3.5.1.3.4.2.9 Servo Performance - The MTT shall not show or indicate any servo period sensitivities under any one reel load condition which will cause malfunction (interrupts) of the tape system when used with the computer; or, which will cause the MTT to fail the design and performance requirements specified herein, when programmed in any one of the following: Forward Start, Reverse Start, Forward Stop, Reverse Stop, at any repetition rate within allowed start/stop times.

3.5.1.3.4.2.10 IRG Length - The MTT shall operate with all legal command sequences with a minimum IRG of 0.688 inch.

3.5.1.3.5 Reel Drive System

3.5.1.3.5.1 Functional Description - The Reel Drive System consists of two independent systems, one for the supply reel and one for the takeup reel. They each drive their respective tape reel in the proper direction and at the required speed to follow the capstan motion and to maintain the tape loops in the proper position within the vacuum columns. Sensing information for these reel servos is obtained from photo senser pickups located within the vacuum column. This information along with other status information from the control system is fed to the reel servo logic circuits and these outputs in turn control the respective tape reel motor amplifiers.

3.5.1.3.5.2 Operational Requirements

3.5.1.3.5.2.1 Reel Hub - See Figure 1 for dimensions. Hub shall be a quick action type for ease of removal of tape reels.

3.5.1.3.5.2.2 Tape Reel - Aluminum reel is shown in Figure 1.

3.5.1.3.5.2.3 Tape Type - Tape shall be in accordance with W-T-0051B, Class I, Size II, except reel shall be as specified herein.

3.5.1.3.5.2.4 File Protect Ring - A plastic insert ring as shown in Figure 2, must be installed on the supply reel to permit writing on tape.

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3.5.1.3.5.2.5 Tape Mark Sensing - Photo cell load point and end-of-tape sensing by reflective strips attached to the tape shall be used. Photo cell low tape sensing through tape load (on supply reel) shall be used.

3.5.1.3.5.2.6 Tape Life - The Magnetic Tape shall withstand at least 10,000 bi-directional (5000 in each direction) passes without damage attributable to the tape transport.

3.5.1.3.6 Data System

3.5.1.3.6.1 Functional Description - the digital data is recorded on tape in a seven track NRZI format. The data system consists of three (3) major sub-systems, the write, the read, and the data control. The data control function is accomplished by the data control logic circuit card. The read and write functions are accomplished by the data read/write card (one for each track) and the magnetic head assembly which is the device which places digital information on and reads it from the magnetic tape.

3.5.1.3.6.2 Operational Requirements

3.5.1.3.6.2.1 Recording Method - NRZI

3.5.1.3.6.2.2 Packing Density - 200, 555.5 and 800 bits per inch.

3.5.1.3.6.2.3 Bad Spot and/or Parity Error Detection - Use record post read method to detect bad spots and/or parity errors.

3.5.1.3.6.2.4 Data Accuracy

3.5.1.3.6.2.4.1 Transient Error Rate - The accuracy of the MTT shall be such that no more than one error occurs in recording or reproducing 10^7 characters under the following conditions:

(1) The test tape has been pretested for dropouts.

(2) Errors which can be localized to a bad spot segment of tape are discounted.

3.5.1.3.6.2.4.2 Permanent Error Rate - The accuracy of the MTT shall be such that no more than one permanent error occurs in recording or reproducing 10^8 characters.

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3.5.1.3.6.2.4.3 Reduced Performance - When operating under the specified reduced performance environment per 3.3.12.1.1.1, the MTT's transient error rate may be increased to 1 in 10^4 characters.

3.5.1.3.6.2.5 Magnetic Head

3.5.1.3.6.2.5.1 Tracks - Seven (7) (6 data, 1 parity).

3.5.1.3.6.2.5.2 Track Spacing - 0.070 inch ± 0.0015 inch.

3.5.1.3.6.2.5.3 Record Head - Gap Length - 0.0010 inch nominal.

3.5.1.3.6.2.5.4 Record Head - Gap Width - 0.048 inch ± 0.0015 inch.

3.5.1.3.6.2.5.5 Read Head - Gap Length - 0.00025 inch nominal.

3.5.1.3.6.2.5.6 Write To Read Gap Spacing - 0.300 inch ± .002 inch.

3.5.1.3.6.2.5.7 Gap Azimuth - 90 degrees plus or minus 1 minute of arc.

3.5.1.3.6.2.5.8 Gap Scatter - Plus or minus 50 micro-inches maximum.

3.5.1.3.6.2.5.9 Total Skew - Less than 5 microseconds at 75 inches per second.

3.5.1.3.6.2.5.10 Erase Head - Gap Width .004 inch min.

3.5.1.3.6.2.5.11 Erase Gap to Write - Gap Spacing - .350 ± .020 inch.

3.5.1.3.6.2.5.12 Record/Post - Read Head - The Record/Post-Read Head subassembly shall be completely interchangeable from one MTT to another in a minimum of time without special tools while still maintaining the electrical and mechanical specifications for the MTT. Azimuth adjustment of the Record/Post - Read Head is permitted.

3.5.1.3.6.2.5.13 Associated Guides and Pressure Packs - All elements shall be securely fastened to a precision base plate. It shall be possible to replace the entire head assembly without exceeding skew limits as specified herein.

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3.5.1.3.6.2.6 Tape Recording Specifications

3.5.1.3.6.2.6.1 Tape Format - The tape format shall be compatible with IBM 727, IBM 729II, and IBM IV magnetic tape transports.

3.5.1.3.6.2.6.2 Recording Method - The recording method shall be NRZI compatible with IBM.

3.5.1.3.7 Pneumatic System

3.5.1.3.7.1 Functional Description - The pneumatic system shall supply a source of vacuum to the vacuum column to provide tape tension for gentle tape handling under all modes of tape motion. Cooling air shall be provided to high heat generating electronic assemblies and a positive pressure shall be provided in the tape path area to assure a clean area for low data error rates. These functions are all performed by motor driven blowers.

3.5.1.3.8 Diagnostic System

3.5.1.3.8.1 Functional Description - The Diagnostic System shall consist of Input Diagnostic Matrix and the Output Diagnostic Matrix circuit assemblies. Additional diagnostic circuitry shall be provided in each system to permit operation of circuits in diagnostic modes which enables an evaluation of circuit outputs for proper operation. When used in conjunction with the computer, the system shall permit MTT evaluation, diagnosis and fault isolation in a rapid manner. Five diagnostic test modes are provided.

3.5.1.4 Controls and Adjustments

3.5.1.4.1 Controls - The maintenance concept employed for this equipment requires organizational level replacement of failed modules. The design shall meet this requirement without adjustment of the replaced assembly or module. The only operational controls and adjustments required are those located on the control panel identified in paragraph 3.5.1.3.3 and the following:

3.5.1.4.2 Door Interlock Switch - This control shall be a three position micro switch assembly which is located on the Low Tape Sensor assembly. It shall be mechanically held in the door closed position when the front door is closed and enables the MTT for "remote" operation by

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the MTC. When the door is open, the switch shall be spring loaded to the center or "door open" position where it disables the reel servos and brakes to allow loading and unloading of the magnetic tape. The third position shall be for local operation of the MTT with the door open and is accomplished by pulling the switch out.

3.5.1.4.3 Record/Post - Read Head Azimuth Adjustment - An adjustment shall be provided to minimize the skew of the record/post - read head to the tape path of the individual MTT. This adjustment shall be self locking and shall remain stable during the period that the head is installed on the MTT:

3.5.1.4.4 Tape Reel Height Adjustment - A means shall be provided to adjust the tape center of the tape reel with respect to the tape path. This adjustment is permitted only when the reel drive units of the MTT are replaced for maintenance purposes. If a guage is required for the adjustment it shall be provided as required by 3.3.7.8.

3.5.1.5 Main Frame Indicators -

3.5.1.5.1 Tape Remaining Indicator - A scale indicating the amount of tape remaining on the supply reel from 1400 to 0 feet shall be provided. The scale shall be visable with the front cover in the closed position.

3.5.1.5.2 Time Totalizing Meters - A dual-time totalizing meter shall be provided and shall not be part of any assembly of the MTT which is normally replaced for maintenance purposes. One meter (MS17322-10) shall indicate the total time during which power has been applied to the MTT. The second meter (MS17322-10) shall indicate the number of hours during which the tape is in motion past the read/write head.

3.5.1.6 Electrical Connections - Connections to external circuits shall be provided as follows:

Reference Designator	Receptacle Type	Function
J1	M81511/01E18-85P1	Input from MTC
J2	M81511/06E16-55P1	Output to MTC
J3	MS3113H14-5P	Power Input

Connectors J1 and J2 must meet the requirements of MIL-C-81511 Class F. Connector J3 must meet the requirements of MIL-C-26482.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements. (These tests are to be performed only when specifically required in the Contract Schedule).

(1) Preproduction (first article) Tests
(2) Initial Production Tests
(3) Acceptance Tests
(4) Life Tests

4.2 Preproduction (First Article) Tests - Preproduction tests shall be made on an equipment representative of the production equipment to be supplied under the contract. Preproduction tests shall be accomplished under the approved test procedure of 4.6. The Government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the procuring activity.

4.2.1 Preproduction (First Article) Test Data - The contractor shall submit all data collected in conducting these tests to the procuring activity for review and approval.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of the specification and the contract, and shall include:

(1) Environmental Tests in accordance with MIL-T-5422 as modified by the requirements of 3.3.10.

(2) Interference Tests in accordance with paragraph 4.2.2.2.

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4.2.2.1 Power Input Tests - An equipment shall be tested to determine that the input power (current, voltage, frequency, etc.) over which it will satisfactorily operate will equal or exceed that specified in paragraph 3.3.12.

4.2.2.2 Interference Tests - Tests shall be conducted in accordance with WR-101, Part I with the following exceptions:

(1) Interference limits shall be as specified in paragraph 3.3.6.

4.2.3 Preproduction (First Article) Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under contract.

4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as approved on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the Government; it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial Production Tests - One of the first ten production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. No other tests shall be conducted on equipment prior to starting the Initial Production Tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction workmanship, performance and quality and that it meets all applicable requirements.

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4.3.2 Accessory Material - In addition to the complete equipment submitted for Initial Production Tests, the contractor shall also submit such accessory material and data as necessary to test the equipment.

4.3.3 Initial Production Sample Approval - Approval of the Initial Production Sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test, the equipment shall be reworked by the contractor by replacing all worn or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing qualitative and quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests
(2) Sampling Tests
(3) Reliability Assurance Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum each equipment accepted shall have passed the following tests:

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(1) Examination of Product

(2) Operational Test

(3) Manufacturing Run In Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated at room ambient long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Run In Test - Each equipment shall be operated under the conditions specified herein for a period of 50 hours (five temperature cycles) in accordance with MIL-STD-781 B, Test Plan XXIX, except that five failures maximum are permitted for each 650 hours of equipment screening test and these 650 hours total shall not accumulate over the period of the contract. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Equipment shall be temperature cycled from -25°C to +55°C portion of the temperature cycle shall be four hours and the +55°C portion shall be 10 hours. The change rate from low to high shall be at 5°C/minute.
Humidity	Dew point shall be less than chamber temperature at all times.
Vibration	Equipment shall be vibrated at 0.036 inches double amplitude for 10 minutes each hour at the +55°C period. Vibration frequency between 20 and 30 Hz.

During each 55°C period, the equipment shall be energized and cycled through its various phases of tape motion. The equipment shall be subjected to an operational test at the conclusion of each temperature cycle. The tape motion cycles and operational tests shall be approved by the procuring activity prior to commencement of testing. Analysis of recurring failures shall be performed and corrective action initiated at the direction of the procuring activity. Test records shall be provided for each equipment.

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4.4.2 Reliability Assurance Tests

4.4.2.1 Reliability tests when required by the procuring activity, shall be conducted in accordance with Military specification MIL-STD-781 except as specified herein, for both the preproduction phase and the production acceptance phase.

4.4.2.2 Test Level - Test Level E of MIL-STD-781 shall be used for both the preproduction phase and the production phase testing except the lower temperature limit shall be -25° centigrade and the vibration shall be 20 to 30 Hz, 0.036 inches double amplitude. The length of the heat portion of the cycle, performance characteristics to be measured and special failure criteria shall be part of the test specifications approved by the procuring activity prior to the commencement of testing.

4.4.2.3 Test Plan - The test plan to be followed in the performance of the reliability tests shall be as follows:

4.4.2.3.1 Preproduction Phase - The preproduction phase test plan shall be in accordance with test plan XX of MIL-STD-781. Two magnetic tape transports shall be used to accumulate test time.

The reliability test shall be continued for a period equal to the specified MTBF (650 hours) multiplied by 3.9 as a total accumulated test time. Six or more failures occurring within the total accumulated test time shall be considered as failure of the reliability test and the basis of rejection.

4.4.2.3.2 Production Acceptance Phase - Same as all Equipment Test, Page 33, paragraph 4.4.2.9.

4.4.2.4 Procedure Following a Rejection Decision - MTT's rejected under the Reliability Test shall be reworked by the contractor at his cost, and resubmitted for test.

4.4.2.5 Preventive Maintenance - No Preventive Maintenance shall be performed during any phase of the reliability testing except when approved by the procuring activity. Any unapproved preventative maintenance shall be considered prima facie evidence of a failure.

4.4.2.6 Failure Definitions

4.4.2.6.1 Failure - The inability of the MTT to perform its required function limits defined herein.

4.4.2.6.2 Independent Failure - A part failure which is not caused by the malfunctioning of associated parts.

4.4.2.7 Non-Chargeable Failures - The following types of failures shall be scored as non-chargeable.

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(1) All MTT failures which do not disable a mission function or cause a false indication of unit malfunction.

(2) The contractor may apply to the procuring activity for approval to eliminate a failure from the reliability computation in either of the two following cases:

(a) The contractor analytically and/or by test establishes the effectiveness of the corrective action (or change) to the satisfaction of the procuring activity, and the corrective action or change is scheduled for introduction and/or retrofit by the contractor as approved by the procuring activity. Approval by the procuring activity shall not protect any subsequent identical failure from classification as relevant.

(b) The contractor demonstrates to the satisfaction of the procuring activity that the failure was a dependent failure.

4.4.2.8 Data Acquisition

4.4.2.8.1 Operating Time Data - Elapsed operating time shall be monitored and recorded in the test log. Time meter readings shall reflect the time during which the MTT's are operated when not under reliability test.

4.4.2.8.2 Failure Data - Failure data will originate in the form of a failure malfunction report which shall include as a minimum all item malfunctions and discrepancies, time meter reading at the time of each malfunction detection, and a detailed history of the resulting maintenance and repair. A detailed failure analysis shall be conducted on the failed part. A corrective action loop shall be established and the elimination of the assignable cause of the failure shall be documented. The effectiveness of the proposed corrective action shall be established by testing.

4.4.2.9 All Equipment Test - Each equipment produced except those submitted for the Reliability Qualification Test, shall be tested for a minimum of 100 hours. Prior to the 100 hour test on each equipment, a burn-in period may be used at the option of the contractor. If the burn-in period is to be used the details thereof must be included in the approved test procedures. The last 50 hours of this All Equipment Test shall be accomplished failure free.

To determine whether the MTBF is being met at any time during the contract the operating test hours and the failures therein (not counting burn-in failures or burn-in operating time) shall be

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totaled and the results compared with the reject line of Test Plan II of MIL-STD-781. (Extend the line as necessary to accomodate the data.) These totals shall accumulate so that at any one time the experience from the beginning of the contract is included. At the conclusion of each month the test results shall be sent to the procuring activity and to the Naval Air Systems Command, Attention: Avionics Division. At any time that the current totals of test hours and test failures plotted on Test Plan II curves show a reject situation, the procuring activity shall be notified. The procuring activity reserves the right to stop the acceptance of equipment at any time that a reject situation exists pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment workmanship, etc., so that the entire compilation will show other than a reject decision.

4.4.3 Special Tests - Special Tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the shakedown or reliability assurance tests.

4.4.3.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate that additional tests are required. (This will be determined by the procuring activity).

4.4.3.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.4 Equipment Failure - Should a failure occur during either the reliability assurance or special tests, the following action shall be taken:

(1) Determine the cause of the failure.

(2) Determine if the failure is an isolated or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occuring in future tests.

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(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life tests. The test shall be 300 hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the Government inspector in accordance with the following. (Equipments which have successfully passed the Initial Production Test, Sampling Tests, Reliability Tests for Special Tests may be selected for life tests.) (Test time accumulated during the Reliability Test may be counted toward the Life Test provided the entire 300 hour is accumulated on a single equipment and equipment selected is in accordance with the table below.)

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 185	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Normal room
Altitude	Normal ground (0-5000 ft.)
Humidity	Room ambient
A. C. Voltage	115 ± 5 volts (at lowest applicable frequency)
D. C. Voltage	27.5 ± 2.0 volts

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

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4.5.3 Performance Check - At approximately eight hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.6 Test Procedures - The procedures used for conducting preproduction tests acceptance tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the Government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.7 Reconditioning of Tested Equipment - Equipment which has been subjected to initial production, acceptance and life tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.8 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government inspector.

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5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specifications MIL-E-17555 and MIL-STD-794. In the event the equipment is not covered in Specification MIL-E-17555, the method of preservation for Level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794.

6. NOTES

6.1 Intended Use - To be used in the P-3C weapon system.

6.2 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number, and date of this specification.
(2) Selection of applicable levels of packaging and packing (see 5.1).

6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.
(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable shall be specifically approved in writing by the procuring activity.
(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

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6.4 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size or weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.5 Type Designations - The type designation may be modified by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.8. The correct type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.6 Revisions - In specification revisions and superseding amendments an asterist " * " preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

6.7 Associated Equipment - MX-8024/AYA-8 Logic Unit 2 (MTC).

*6.8 This Specification is under the cognizance of AIR-53302E

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APPENDIX I

INPUT AMPLIFIER CIRCUIT CHARACTERISTICS

1. SCOPE

1.1 This Appendix defines the characteristics for the Input Amplifier circuit referenced in this Specification.

2. REQUIREMENTS

2.1 Voltage Level Inputs - The Logic "1" state shall be 0 volts + 0.5 volt and the logic "0" state shall be 2.5 to 5.0 volts as measured at the input amplifier. If both inputs of the input amplifier are disconnected this shall be considered a logic "0".

2.2 Input Noise Rejection - The circuit shall be capable of handling common mode voltages of +5 volts peak at the input terminals with no change in output level.

2.3 Input Impedance - The AC input impedance shall be 130 ohms nominal and the DC input impedance shall be 2000 ohms nominal.

2.4 Voltage Level Outputs - The logic "1" state shall be 4.0 to 5.5 volts and the logic "0" state shall be 0 volts + 0.5 volt.

2.5 Output Rise and Fall Times - The time required for the input amplifier circuit to switch logic levels, as measured from the 10% to 90% amplitude points at the output terminals of the amplifier shall be less than 0.5 microsecond when loaded as described in paragraph 2.7.

2.6 Output Propagation Delay - The propagation delay when loaded as described in paragraph 2.7 shall be less than 200 ns.

2.7 Output Loading - The circuit shall be capable of sinking to ground at least 10 milliamperes of current and driving a capacitive load of 50 pf at both output terminals.

2.8 Environmental Performance - The circuit shall operate within performance limits when subjected to temperatures over the range of -55°C to 71°C.

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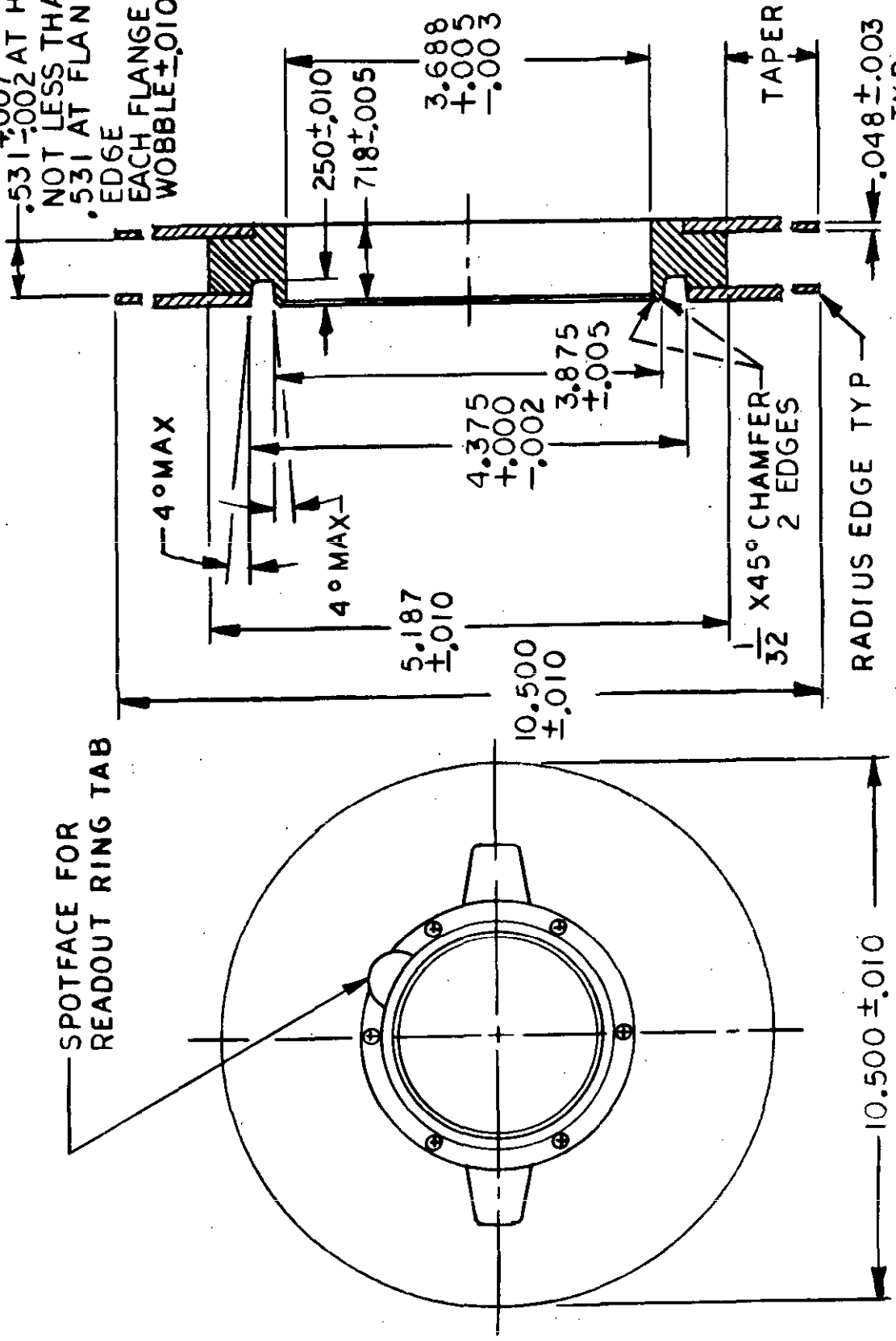


FIGURE 1, MAGNETIC TAPE TRANSPORT TAPE REEL DIMENSION

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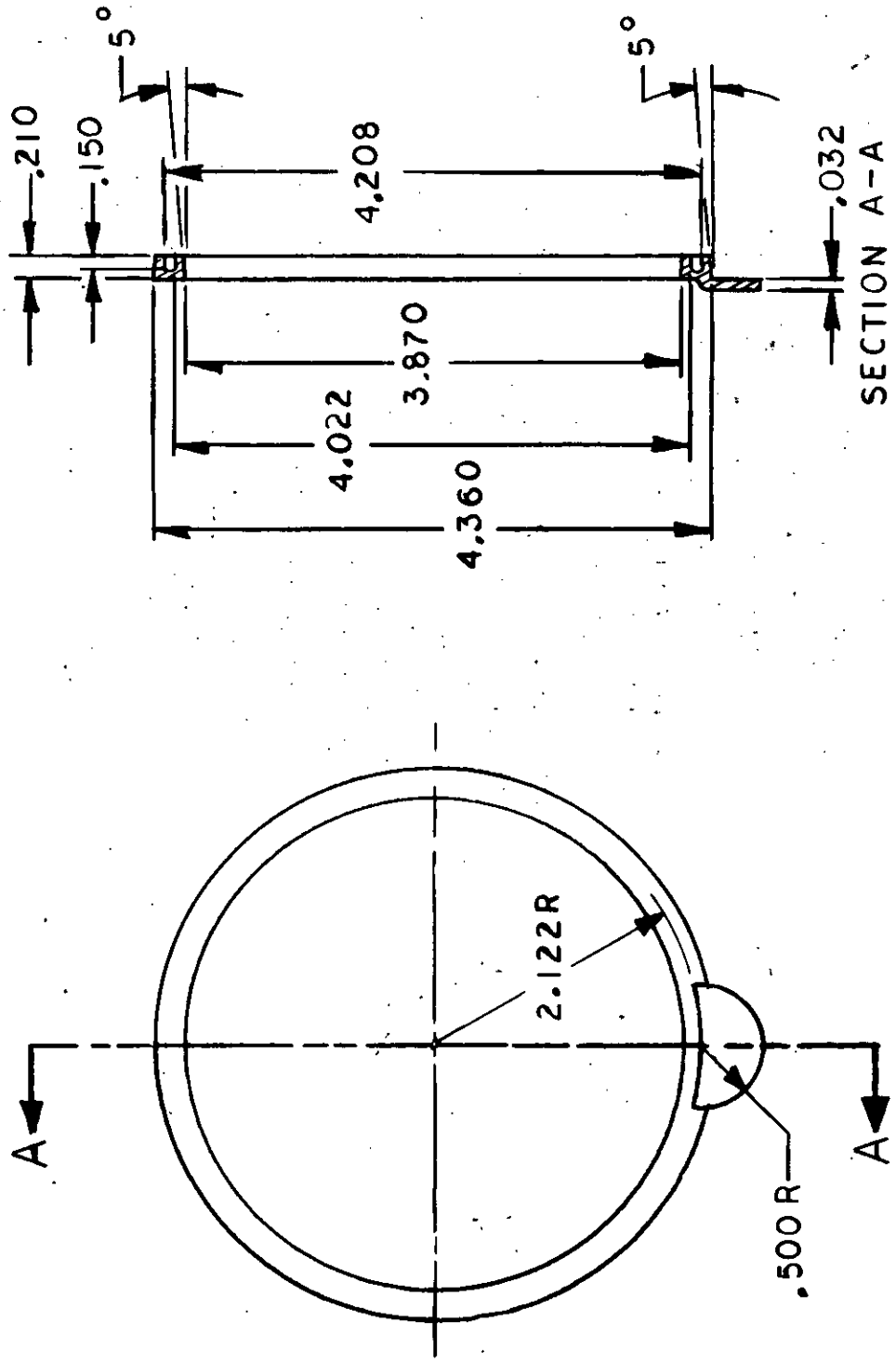


FIGURE 2. MAGNETIC TAPE TRANSPORT TAPE REEL FILE PROTECT RING

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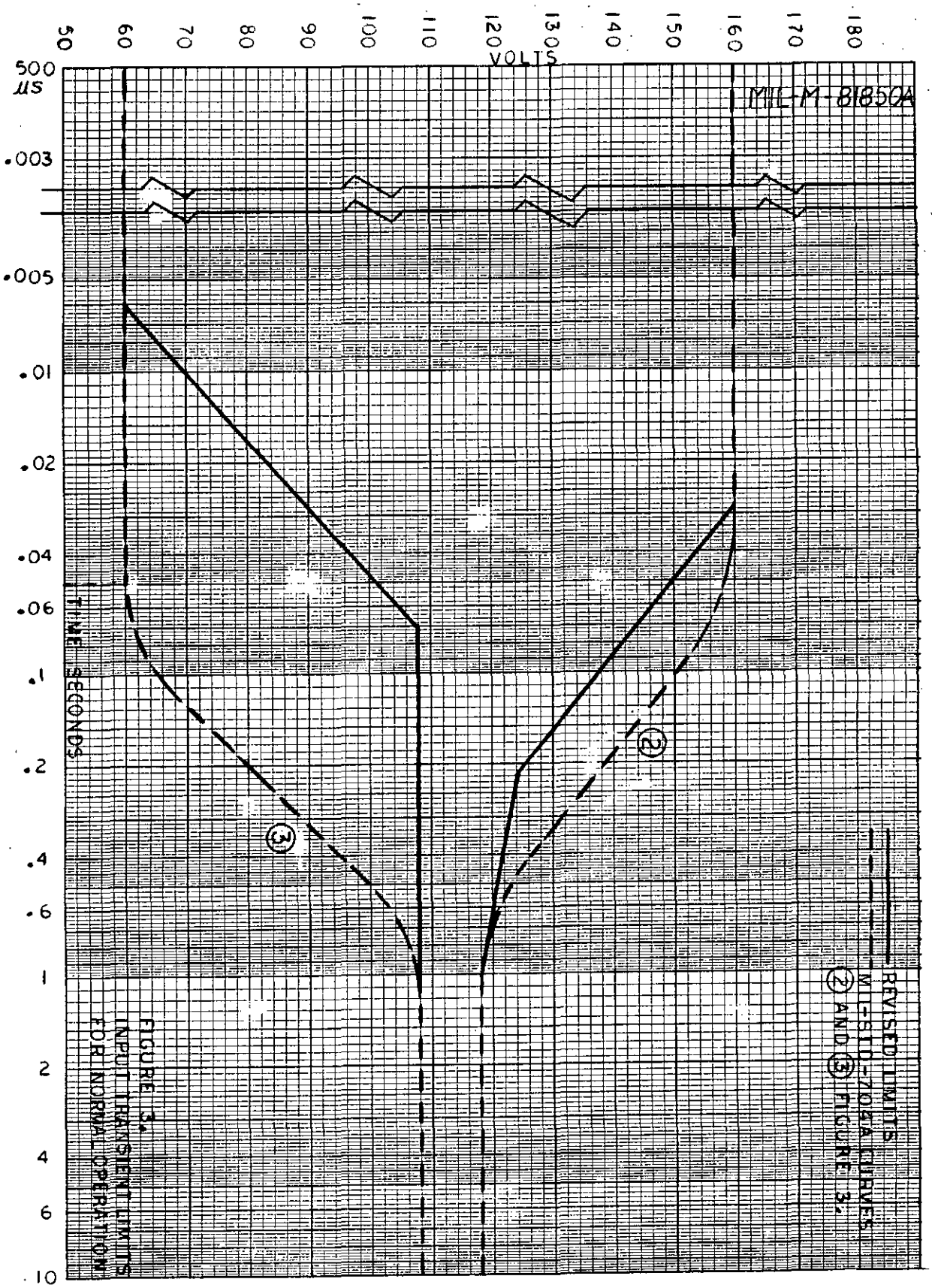


FIGURE 3. INPUT TRANSIENT LIMITS FOR NORMAL OPERATION

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DOCUMENT IDENTIFIER AND TITLE
MIL-M-81850A(AS) MAGNETIC TAPE TRANSPORT RD-319A/AYA-8

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

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